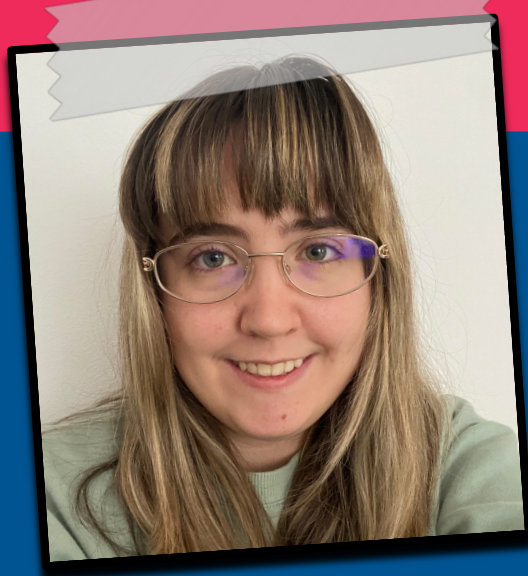


PREDICTING SPIN-DEPENDENT PHONON BANDS WITH MOMENT TENSOR POTENTIALS



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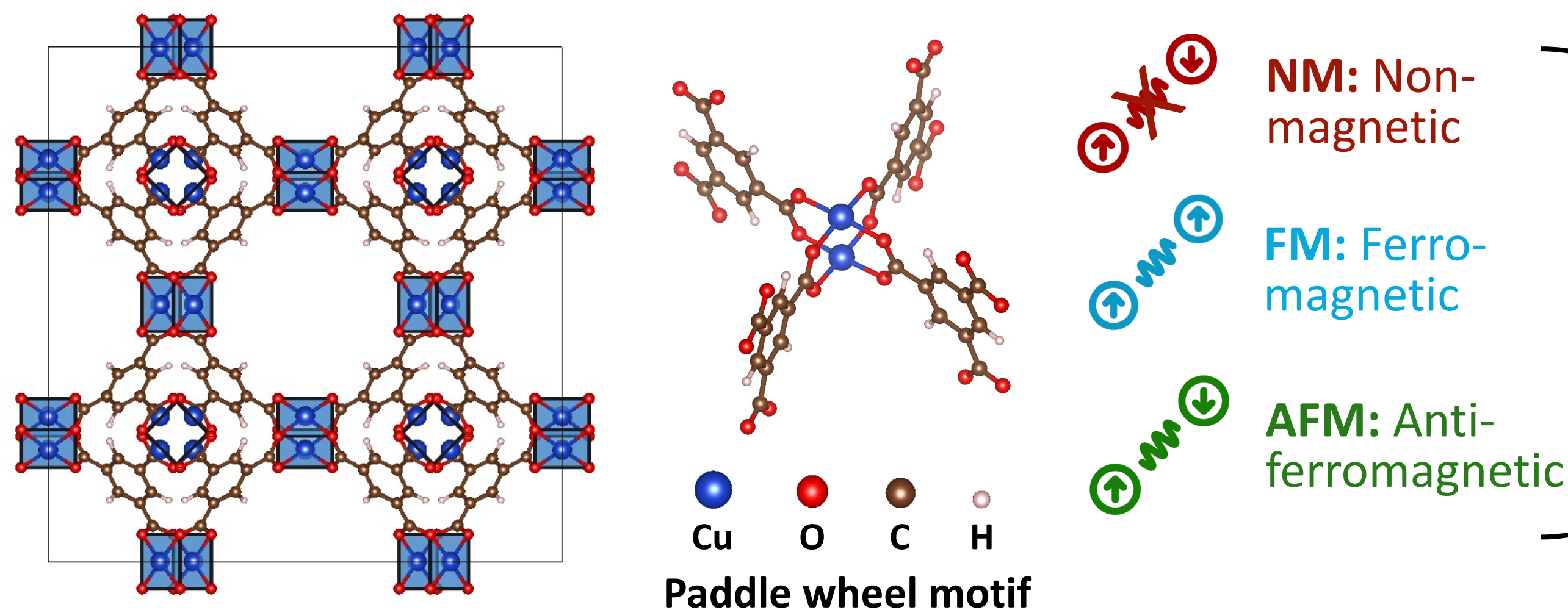
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Full paper is available here!



INTRODUCTION & RESEARCH QUESTION

Metal-organic framework HKUST-1:



In crystalline materials, atoms vibrate collectively. These quantized lattice vibrations are called phonons.

Phonons determine thermal and mechanical properties of solid-state materials.

Cu dimers in HKUST-1 exhibit three different magnetic configurations.

QUESTION #1:

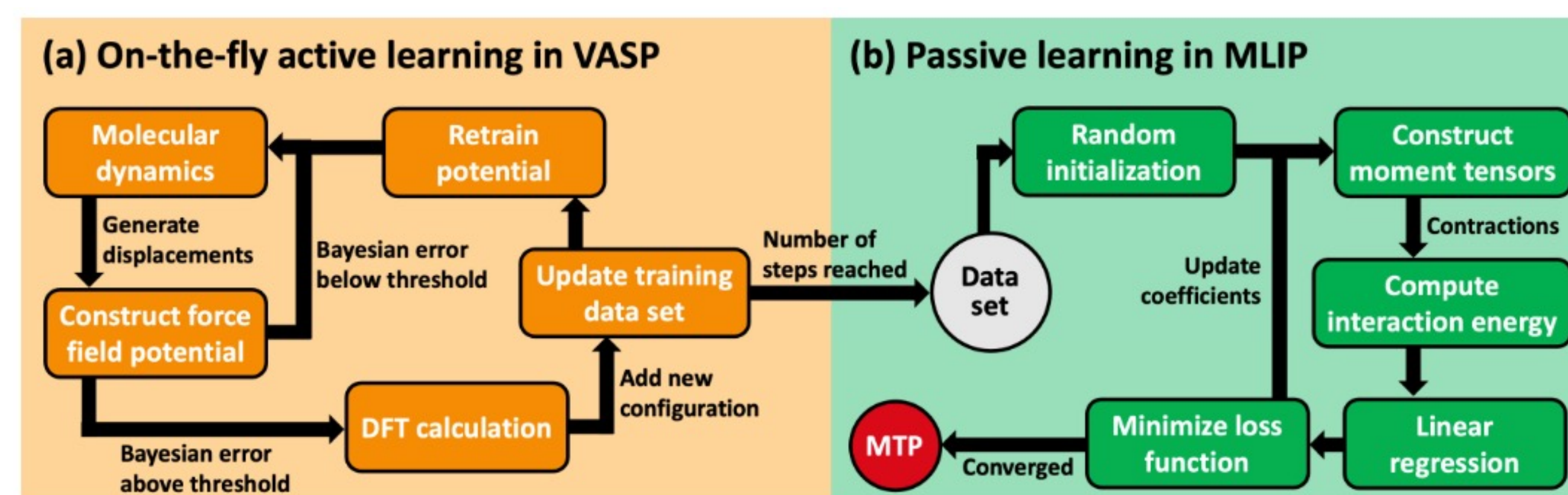
Do different spin states alter the phonon band structure of HKUST-1?

QUESTION #2:

Can this effect be reproduced using spin-specific moment tensor potentials (MTPs)?

COMPUTATIONAL WORKFLOW

- DFT reference calculations
- Generation of training data
- PBE+D3 level of theory
- NPT, heating from 100 K to 500 K
- 50,000 steps, time step of 1 fs
- Active learning allows sampling of a large configurational space



The workflow was repeated three times (one for each spin state).

- Data set: 500 configurations
- Energies, stresses and forces are required
- Each spin-sensitive MTP was trained 5x
- MTP with the lowest loss function was used for further analysis
- MTPs enable the computation of phonons at drastically reduced cost

RESULTS & DISCUSSION

ANSWER TO QUESTION #1:

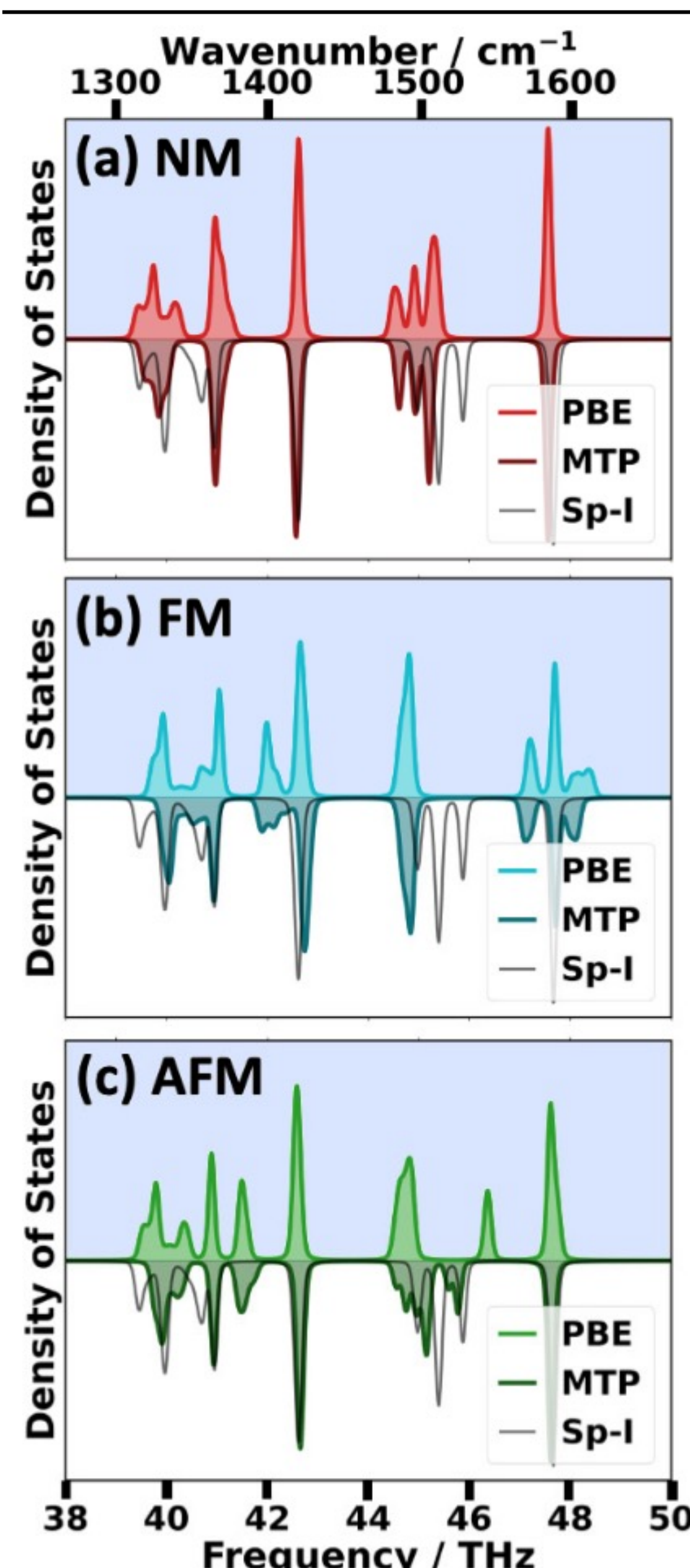
Spin-states affect the phonon spectra!

- Acoustic phonon modes (0–3 THz) show only minor spin dependence.
- Strongest deviations appear in the 38–50 THz region.

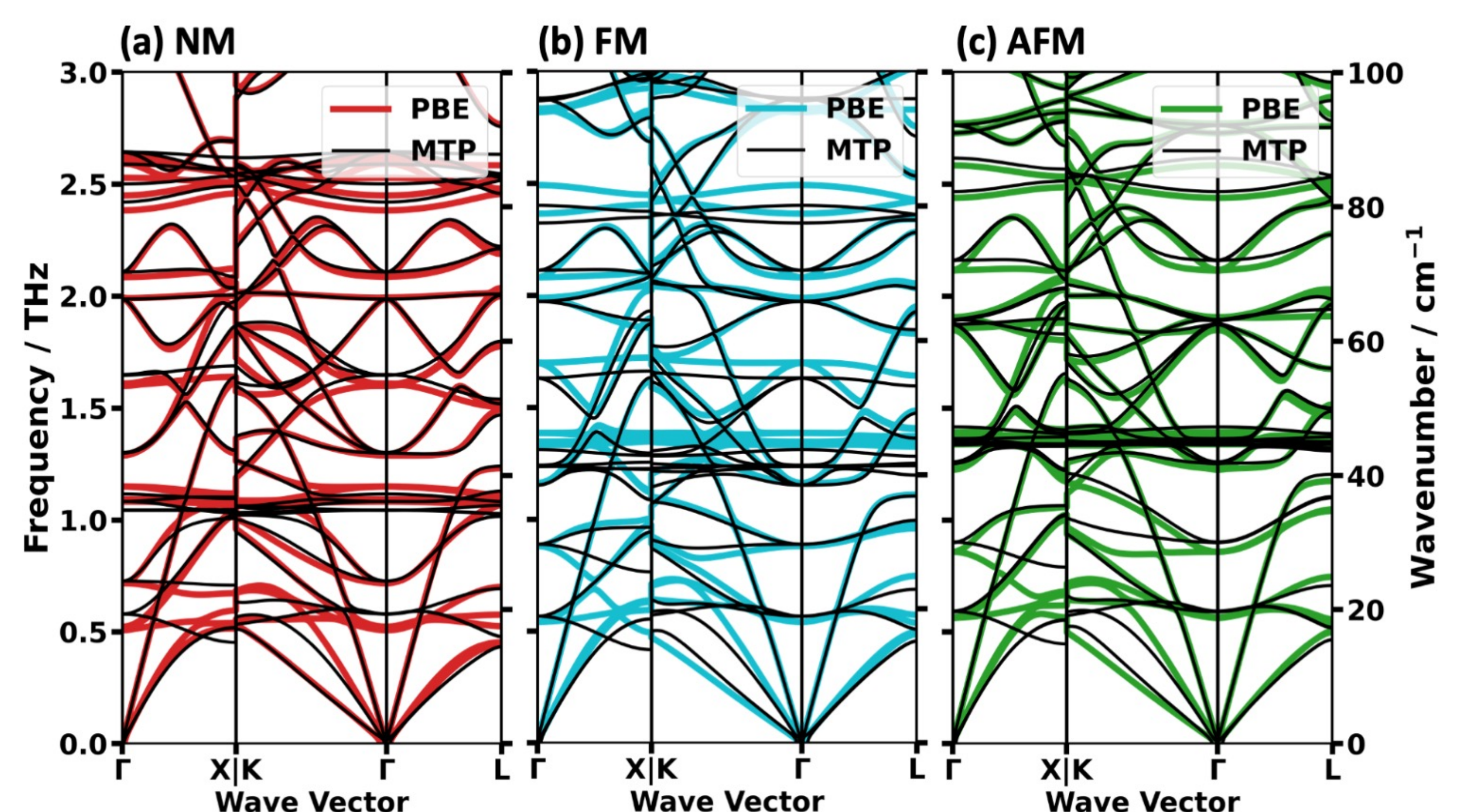
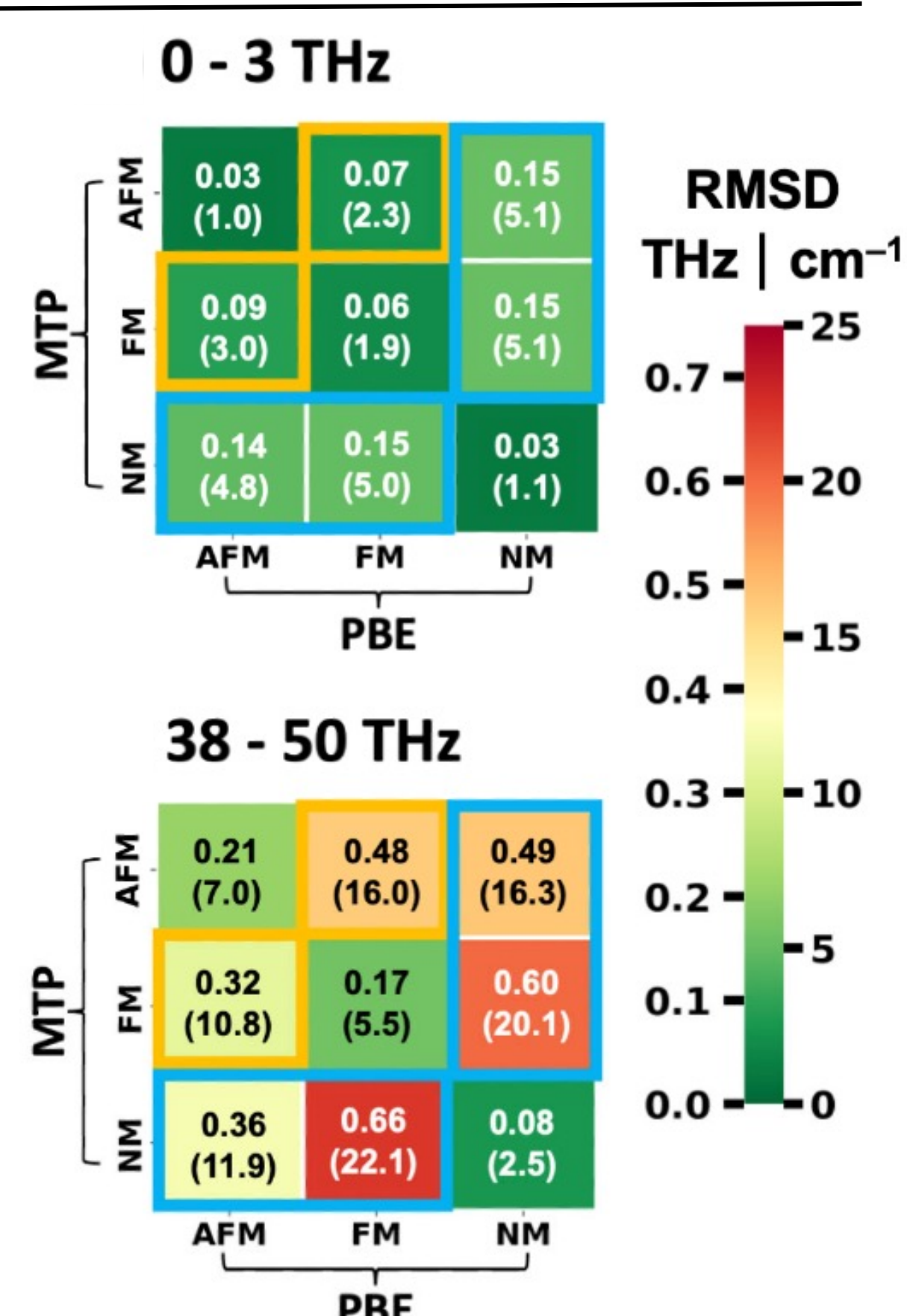
ANSWER TO QUESTION #2:

Spin-specific MTPs accurately reproduce DFT phonon bands!

- Excellent agreement across all magnetic states.
- Low RMSD values (up to 0.2 THz) confirm quantitative agreement.



- Sp-I: spin-ignorant MTP, trained on the entire data set (all states)
- Sp-I represents an 'averaged spin' and is unable to predict the overall DOS shape of any magnetic state



CONCLUSION & OUTLOOK

- Spin polarization alters interatomic force constants governing optical phonons (especially in the region from 38 to 50 THz)
- Excellent quantitative agreement between DFT and spin-sensitive MTPs is achieved in this study
- Magnetic effects are preserved in our machine-learned potentials
- MTPs enable large-scale phonon calculations beyond DFT limitations
- Open the door to finite-temperature lattice dynamics studies
- Allow investigation of thermal transport in large MOF supercells

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ACKNOWLEDGEMENT

This research was funded by the Graz University of Technology through the Lead Project (PMWS-P03). Moreover, this research was also funded by the Austrian Science Fund (FWF) [10.55776/P34463].